

Stop Asking Which Class a Node Belongs To: A Cheaper Question Makes Graph Active Learning Both Better and Cheaper

Peking University and Apple introduce IGP, rethinking active learning for graph neural networks with relaxed queries and soft labels

Graph neural networks (GNNs) are strong at tasks like node classification, but that strength comes with a bill: they need a lot of labeled nodes, and human annotation is slow and expensive. Active learning is the usual way to cut that bill. Instead of labeling blindly, each round picks the nodes most worth labeling and sends only those to an expert. Yet no matter how clever the selection strategy is, almost every prior graph active-learning method rests on the same unspoken assumption: that an expert can always name the exact class a node belongs to.

That assumption breaks down when the label space is large or the domain is unfamiliar. Take ogbn-papers100M from the Open Graph Benchmark, which has 172 classes. Asking an annotator to pinpoint the exact class of a node out of a hundred-plus candidates is beyond what most people can do reliably, and the labeling budget gets spent on questions that are simply too hard to answer well.

This ICLR 2022 paper takes a different route. Rather than asking "which class does it belong to?", it asks a much cheaper binary question: "does it belong to the (model's guessed) first category, yes or no?" This relaxed query does not return an exact class; it returns a soft label. Paired with it is a new selection criterion that deliberately picks the nodes whose information gain propagates farthest across the graph. The result: higher accuracy at the same labeling cost.

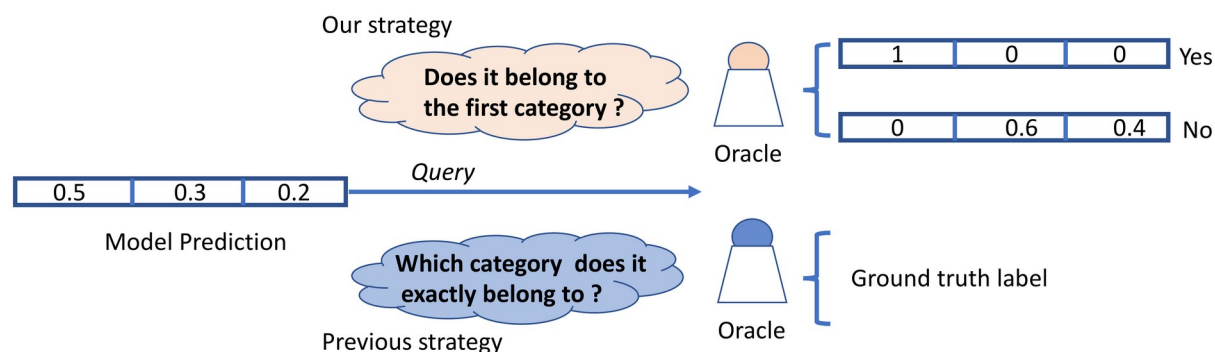


Figure 1. Two labeling strategies. Top, the paper's relaxed query: given the model's prediction (0.5/0.3/0.2), the oracle only answers "does it belong to the first category?", and the yes/no reply becomes a normalized soft label (yes -> 1/0/0, no -> 0/0.6/0.4). Bottom, prior work: the oracle must name the exact ground-truth class. The binary question on the left is plainly easier to answer than the multi-class question on the right.

Paper: Information Gain Propagation: A New Way to Graph Active Learning with Soft Labels (ICLR 2022, arXiv:2203.01093, <https://arxiv.org/abs/2203.01093>). The authors are from Peking University and Apple.

The problem: the expensive part is also the wrong question

Break the labeling cost apart and two long-overlooked sources of waste appear. The first is the cost of the query itself. Asking an expert to identify the exact class among c categories is roughly $c-1$ times as demanding, and as expensive, as answering a single binary question, and the gap grows with the number of classes. Prior methods implicitly treat one exact annotation as the atomic unit of cost, so their budgets are inflated from the start.

The second is the narrow view of the selection criterion. Most prior criteria were designed for hard labels and measure only the uncertainty of a single node in isolation. They ignore a defining property of GNNs: labeling one node propagates supervision along the graph to its k -hop neighbors. A node's real value is therefore not just how uncertain it is on its own, but how much of its neighborhood it can illuminate.

The method: let information gain propagate through the graph

IGP (Information Gain Propagation) folds both ideas into a single loop. The pipeline cycles through three stages: train a GNN on the currently labeled nodes using their soft labels (Model Training); pick the nodes most worth labeling this round with the new criterion (Node Selection); and issue relaxed queries on the chosen nodes, turning each yes/no answer into a normalized soft label before updating the data (Node Labeling). Then it repeats.

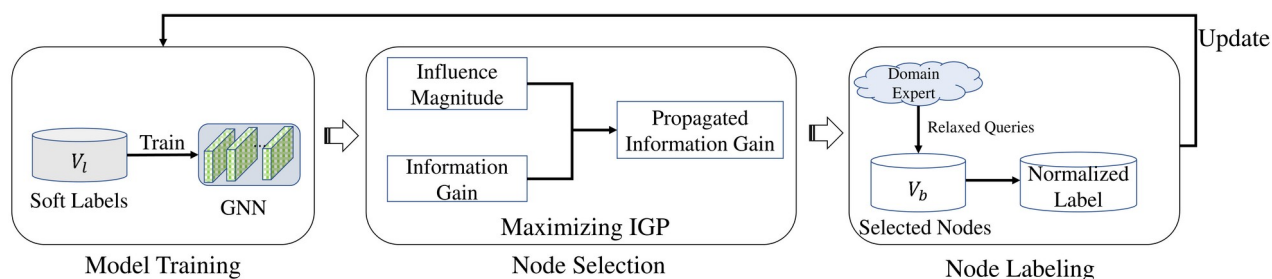


Figure 3. Overview of the IGP framework. Left: a GNN is trained on the labeled set V_l with soft labels. Center: selection combines each node's Influence Magnitude with its Information Gain, whose product is the Propagated Information Gain. Right: relaxed queries on the selected set V_b yield normalized soft labels, which feed back to update the model.

The selection criterion is the heart of IGP and where it parts ways with earlier work. For each candidate node, IGP does not just ask how much labeling it would reduce that node's own uncertainty; it asks how much uncertainty would fall across every node in its receptive field once the label propagates. Concretely, it estimates a node's influence magnitude on its neighbors via feature-gradient / random-walk propagation over a k -layer GNN, then weights each influenced node's information gain, its drop in entropy, by that influence. Summing the product over the receptive field gives the node's information-gain-propagation score.

Selection then becomes a budget-constrained maximization: within a labeling budget B , choose the subset of nodes that maximizes the total propagated information gain across the graph. In plain terms, IGP does not pick the node that is most confused about itself; it picks the node whose single label lights up the most of its neighborhood.

Results: a consistent lead across five datasets and four backbones

The authors evaluate on five standard graph benchmarks: the citation networks Cora, Citeseer, and PubMed, the large social network Reddit, and ogbn-arxiv from the Open Graph Benchmark. Crucially, the budget is not counted as the number of nodes labeled but as the true annotation cost, since an exact multi-class query is $c-1$ times more expensive than a binary one. Under an equal budget of 20 labels per class with a GCN backbone, IGP reaches the highest accuracy on all five datasets.

Table 1. Test accuracy (%) under an equal labeling budget (20 labels per class, GCN backbone).
Against the strongest baseline, GRAIN, IGP gains 1.6-2.2% on the citation networks, +0.9% on Reddit, and +0.6% on ogbn-arxiv.

Method	Cora	Citeseer	PubMed	Reddit	ogbn-arxiv
Random	78.8	70.8	78.9	91.1	68.2
ALG	83.6	73.6	80.9	92.4	70.1
GRAIN	84.2	74.2	81.8	92.5	70.3
IGP	86.4	75.8	83.6	93.4	70.9

More telling than any single number is how accuracy scales with budget. As the labeling budget grows, IGP's accuracy climbs the fastest and stays ahead at every budget point. The advantage is not an artifact of one lucky budget setting; it holds across the whole curve.

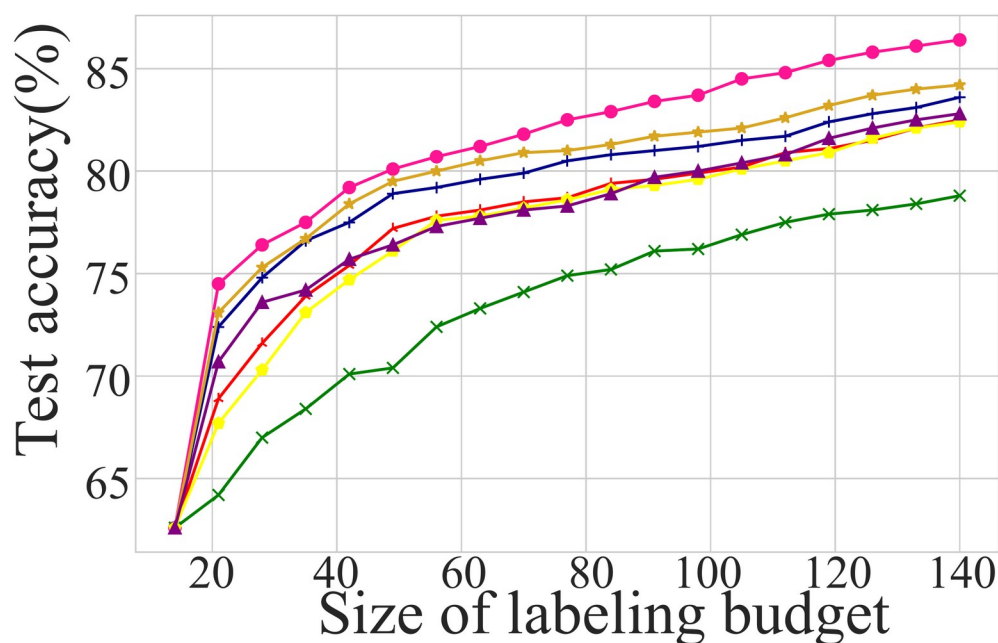


Figure 4. Test accuracy as the labeling budget grows (citation network, GCN). The top pink curve is IGP: it leads from the smallest budget onward, and the margin over the other active-learning baselines widens as the budget increases.

The lead is robust to the choice of backbone as well. On PubMed, across SGC, APPNP, GCN, and MVGRL, IGP beats both ALG and GRAIN by more than 1.8%, which suggests the payoff from relaxed queries plus propagation-aware selection does not hinge on any one network architecture.

Why it works: more spread-out selections, and every piece pulls its weight

What makes IGP stronger? One intuition comes from where its chosen nodes land on the graph. Projecting the nodes IGP selects on Cora to two dimensions with t-SNE shows them spread fairly evenly across the embedding space rather than clustering in a few regions. A propagation-aware criterion naturally favors nodes whose influence covers different neighborhoods, so the resulting labeled set is more representative and less redundant.

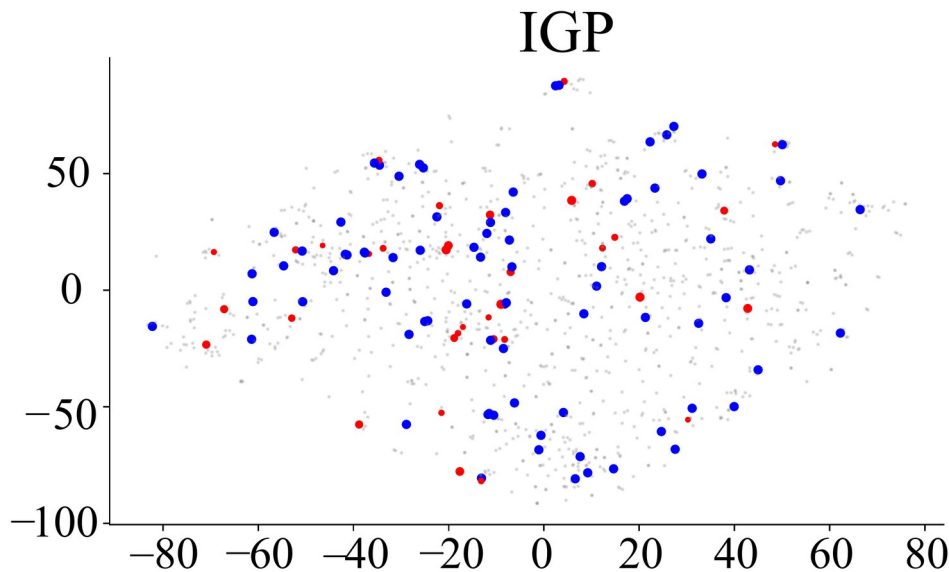


Figure 5. t-SNE distribution of the nodes IGP selects on Cora. Colored points are the selected nodes; gray points are all nodes. The selections spread across the entire embedding space, covering distinct regions, which reflects the diversity and representativeness of the selection.

An ablation confirms that each component earns its place: removing any one of them costs accuracy. Dropping the propagation-aware selection (informative selection) hurts most, up to -3.2% on PubMed and -2.5% on Cora, making it the single most important ingredient. Removing soft-label training (informative training) costs up to -2.9%, dropping the influence magnitude (information quantity) up to -2.1%, and removing the normalized soft label up to -2.4%. The reading is clear: what really separates IGP from the pack is propagation-aware selection, not simply the switch to soft labels.

Takeaway: change the question, and change how you choose

IGP's recipe is compact: replace the expensive exact-class annotation with a cheap binary yes/no that yields a soft label, and replace the "who is most uncertain" selection rule with "whose information gain propagates farthest across the graph." Together they make graph active learning both more accurate and cheaper to label at an equal cost, and the recipe drops onto any GNN backbone.

It is worth being honest about the boundaries. The soft label is built from the model's current prediction plus a single binary confirmation from the oracle, so its quality depends on how good

that prediction distribution is; the cost advantage rests on the assumption that a binary query really is $c-1$ times cheaper than a multi-class one, and the actual savings will vary with the task and the annotation setup. Still, as a plug-in paradigm, relaxed queries plus propagation-aware selection point graph active learning in a direction worth pursuing.